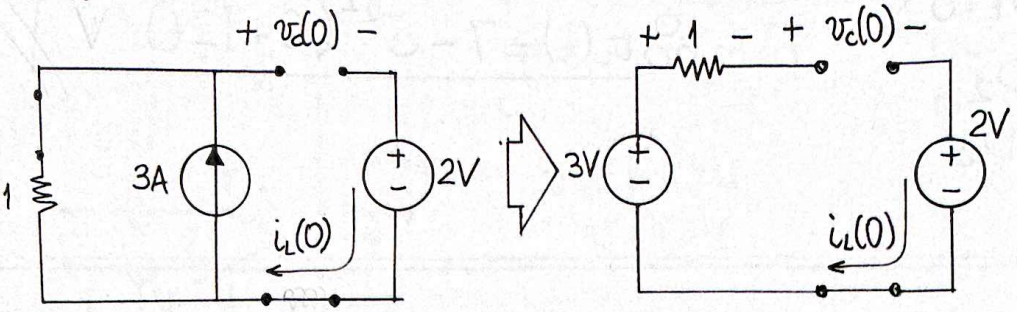
Pregunta 1

① Para $t<0s$:



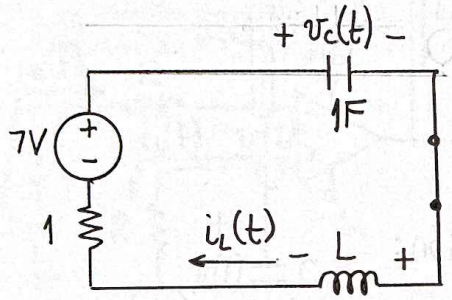
①a Por simple inspección:

$$i_L(0)=0A$$

①b LTK

$$3-1\cdot i_L(0)-v_c(0)-2=0$$
$$1-v_c(0)=0$$
$$v_c(0)=1V$$

② Para $t\geq 0s$:



②a En este caso:

$$i_L(t)=1\cdot\frac{d}{dt}v_c(t)$$
$$i_L(t)=\frac{d}{dt}v_c(t)\ \ldots(1)$$

②b LTK:

$$7-v_c(t)-L\frac{d}{dt}i_L(t)-1\cdot i_L(t)=0$$
$$L\frac{d}{dt}\left(\frac{d}{dt}v_c(t)\right)+\left(\frac{d}{dt}v_c(t)\right)+v_c(t)=7$$
$$L\frac{d^2}{dt^2}v_c(t)+\frac{d}{dt}v_c(t)+v_c(t)=7$$
$$\Rightarrow v_c(\infty)=v_f(t)=7V$$

②c Ecuación característica:

$$Ls^2+s+1=0$$
$$\Rightarrow s_{1,2}=\frac{-1\pm\sqrt{1^2-4(L)(1)}}{2(L)}$$

Como $s_{1,2}$ corresponde a una respuesta Críticamente amortiguada:

$$1^2-4(L)(1)=0$$
$$\Rightarrow \therefore L=0.25\,H$$

②d Como la respuesta es Críticamente Amortiguada:

$$s_{1,2}=\frac{-1}{2(L)}=\frac{-1}{2(0.25)}=-2\ rad/s$$
$$\Rightarrow v_{cn}(t)=e^{-2t}(D_1+D_2t)$$
$$\Rightarrow v_c(t)=v_{cn}(t)+v_f(t)$$
$$v_c(t)=e^{-2t}(D_1+D_2t)+7\ldots(2)$$

②e Para $t=0$ en (2):

$$v_c(0)=e^{-2(0)}(D_1+D_2(0))+7$$
$$1=D_1+7$$
$$D_1=-6V$$

(2f) Para $t=0s$ en (1):

$$i_L(0) = \frac{d}{dt} v_c(t)\Big]_{t=0}$$

$$i_L(0) = -2D_1 + D_2$$

$$(0) = -2(-6) + D_2$$

$$D_2 = -12\,V/s$$

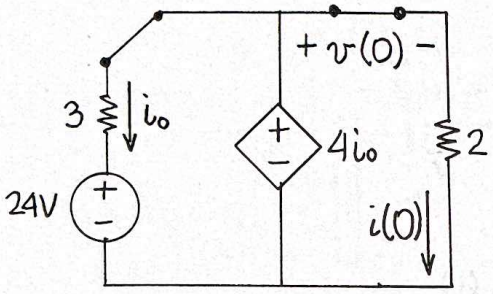
(2g) Entonces; en (2):

$$v_c(t) = e^{-2t}((-6)+(-12)t) + 7$$

$$\therefore v_c(t) = 7 - e^{-2t}(6+12t)\ V$$

Pregunta 2

(1) Para $t<0s$:



(1a) LTK:

$$24 + 3i_o - 4i_o = 0$$

$$i_o = 24A$$

(1b) Por simple inspección:

$$v(0) = 0V$$

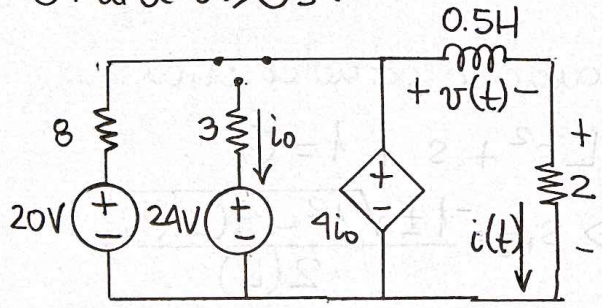
(1c) LTK:

$$4i_o - v(0) - 2i(0) = 0$$

$$4(24) - (0) = 2i(0)$$

$$i(0) = 48A$$

(2) Para $t\geq 0s$:



(2a) Por simple inspección:

$$i_o = 0A$$

(2b) LTK:

$$4i_o - v(t) - 2i(t) = 0$$

$$4(0) - 0.5\frac{d}{dt}i(t) - 2i(t) = 0$$

$$0.25\frac{d}{dt}i(t) + i(t) = 0$$

$$\Rightarrow i(t) = i(0)\,e^{-t/(0.25)}$$

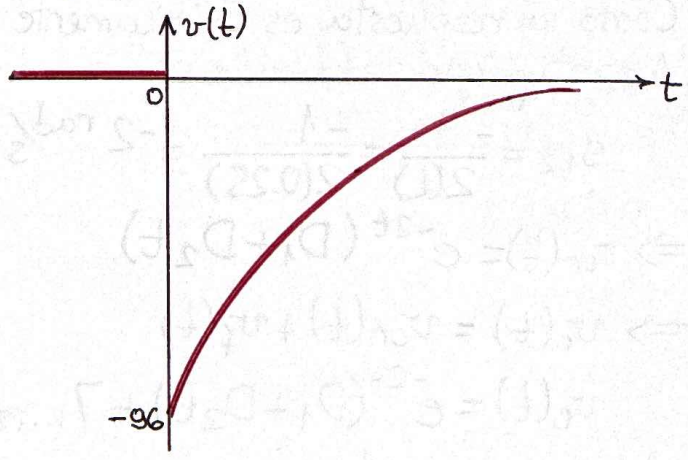
$$i(t) = 48e^{-4t}\ A$$

(3) Para $v(t)$:

$$v(t) = 0.5\frac{d}{dt}i(t)$$

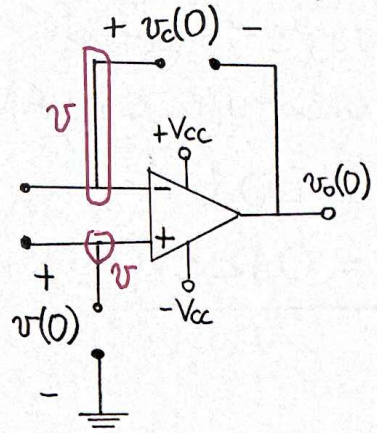
$$\therefore v(t) = -96e^{-4t}\ V$$

(4) Graficando $v(t)$:



Pregunta 3 parte a)

① Para $t<0s$:



①a Como los capacitores no tienen carga:

$v(0)=0V$

$v=0V$

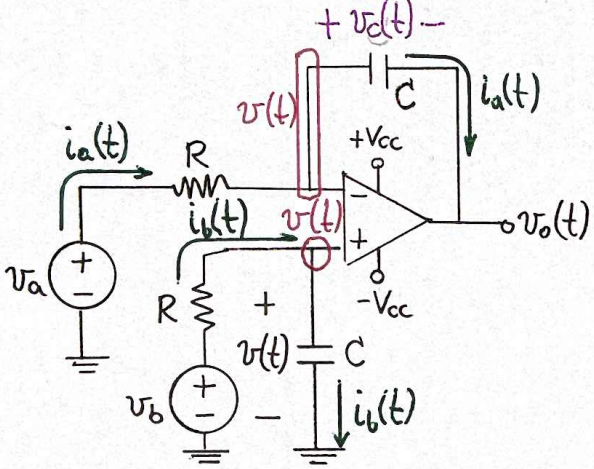
①b Entonces, como los capacitores no tienen carga:

$v_c(0)=v-v_o(0)$

$(0)=(0)-v_o(0)$

$v_o(0)=0V$

② Para $t\geq 0s$:



②a LTK:

$$v_b - Ri_b(t) - v(t) = 0$$

$$Ri_b(t) + v(t) = v_b$$

$$RC\frac{d}{dt}v(t) + v(t) = v_b$$

$$\Rightarrow v(t) = v_b + [v(0) - v_b]e^{-t/RC}$$

$$v(t) = v_b - v_b e^{-t/RC} \quad \ldots(1)$$

②b Para $i_a(t)$; usando (1):

$$i_a(t) = \frac{v_a - v(t)}{R}$$

$$i_a(t) = \frac{v_a - v_b + v_b e^{-t/RC}}{R} \quad \ldots(2)$$

②c Usando (1) y (2):

$$i_a(t) = C\frac{d}{dt}(v(t) - v_o(t))$$

$$v_a - v_b + v_b e^{-t/RC} = RC\frac{d}{dt}v(t) - RC\frac{d}{dt}v_o(t)$$

$$v_a - v_b + v_b e^{-t/RC} = v_b e^{-t/RC} - RC\frac{d}{dt}v_o(t)$$

$$\frac{d}{dt}v_o(t) = \frac{v_b - v_a}{RC}$$

$$\Rightarrow v_o(t) = \frac{v_b - v_a}{RC}t$$

Pregunta 3 parte b)

① Usando los datos:

$$v_2(t) = \frac{(0.015)-(0.04)}{(50k)(10n)}\,t$$

$$v_o(t) = -50t \quad \ldots (3)$$

② Como (3) tiene signo negativo; entonces el circuito satura en -6V:

$$(-6) = -50t$$

$$\therefore t = 0.12\,s$$